Mathematical Foundations of Future Convergence Causal Theory

— Unified Formalization of $\Lambda(T)$, Double Fixed-Point Theory, Triple-Temporal Simultaneity, and the Semantic Density Tensor —

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Abstract

The purpose of this paper is to provide a unified formalization of the core mathematical objects that repeatedly appear in Future Convergence Theory (FCT):

$\Lambda(T)$, double fixed-point structure, triple-temporal simultaneity, semantic density tensor, quantum creative allowance

and to place them within a single coherent framework.

Here, concepts that have been introduced separately across various applied papers (cosmology, quantum theory, theories of consciousness, AI, language, international relations, and so on) are re-positioned on top of a more abstract yet explicit mathematical structure. This paper does not aim at a fully axiomatized system. Rather, it is positioned as an entry point to the foundational mathematics of FCT, whose role is to demonstrate that *"all applications can in principle be reconstructed from a single underlying structural core."*

In particular, we show that:

- time is redefined as a "double fixed-point structure" based on future fixed points together with present/past fixed points;
- the physical world, information, consciousness, society, and civilization are described as *dynamics of semantic density tensors* defined on a common abstract information field $\mathcal{M}$;
- quantum phenomena are formalized as a "creative allowance" that bridges a four-dimensional future information field and three-dimensional manifestation.

Through this, we clarify how a principled departure from classical past-causal theories and observation-dependent "quantum civilization" pictures can be described within a single mathematical scheme.

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1 Introduction: Objectives and Basic Position

The goal of this paper is to concentrate within a single document the *core mathematical objects* that are scattered throughout various papers in Future Convergence Theory (FCT), and to present them as a "mathematical core" with their mutual relationships made explicit.

The basic position adopted in this paper is as follows.

1. Temporal structure is not a linear series of past $\to$ present $\to$ future, but rather a *double fixed-point structure* characterized by future fixed points together with present/past fixed points.

2. The physical world, information, consciousness, society, and civilization are all described as *dynamics of semantic density tensors* defined on a common abstract information field $\mathcal{M}$.

3. The quantum layer is an intermediate layer *located between* a four-dimensional future information field $\Phi(T)$ and a three-dimensional manifestation $X(t)$, and is characterized by a "creative allowance" as its admissible structure.

In what follows, we introduce only the minimally necessary mathematical objects and show that more sophisticated applications (cosmology, theories of consciousness, AI, language, international relations, etc.) can *in principle be reconstructed from this core alone.*

2 Temporal Structure and the Future Convergence Function $\Lambda(T)$

2.1 Time Parameter and State Space

We first introduce a time parameter $t \in \mathbb{R}$ and let the state space $\mathcal{X}$ be the set of all "manifestable states of the world." We treat $\mathcal{X}$ as an abstract topological or measure space.

- $x(t) \in \mathcal{X}$: the "current state" as seen from a given observational system;
- $\mathcal{H}$: a Hilbert-space representation of the states (quantum or informational representation).

In this paper, we do not assume the "flow" of time. Instead, we redefine time as a constraint coming from future fixed points, as will be described below.

2.2 Future Fixed Points and the Future Convergence Function

The most basic object in FCT is the future fixed point.

Definition 2.1 (Future fixed point)**.** A point $x^\star \in \mathcal{X}$ on the state space $\mathcal{X}$ is called a *future fixed point* if there exists a sequence of functions (or a dynamics) $F_T : \mathcal{X} \to \mathcal{X}$ such that, for $T \to +\infty$,

$$\lim_{T\to+\infty} F_T(x_0) = x^\star$$

holds for a broad class of initial states x_0.

Here, T is a "scale parameter on the future side" and is distinguished from the ordinary time parameter t. Intuitively, T indicates "how deeply one looks into the convergence structure on the future side."

Definition 2.2 (Future convergence function $\Lambda(T)$)**.** We define the future convergence function $\Lambda(T)$ as a family of probability measures (or weights) on the state space $\mathcal{X}$:

$$\Lambda : \mathbb{R}_+ \to \mathcal{P}(\mathcal{X}), \qquad T \mapsto \Lambda(T),$$

where $\mathcal{P}(\mathcal{X})$ denotes the space of probability measures on $\mathcal{X}$. $\Lambda(T)$ expresses "to what extent each state $x \in \mathcal{X}$ is preferred at the future scale T."

The larger T becomes, the higher the weight of the states near the future fixed point grows. In the idealized limit $T \to \infty$, the weight concentrates in a delta-like way on the future fixed point (or set thereof).

Definition 2.3 (Convergence strength)**.** For a state $x \in \mathcal{X}$, the convergence strength $\kappa(x)$ is defined by

$$\kappa(x) := \lim_{T\to\infty} \frac{1}{T} \log \Lambda(T)(U_x),$$

where U_x is a sufficiently small neighborhood containing x. $\kappa(x)$ expresses to what extent that state is exponentially preferred as one extends the scale on the future side.

Intuitively, the larger $\kappa(x)$ is, the more that state becomes a "high-necessity structure from the viewpoint of the future."

3 Double Fixed-Point Theory and Triple-Temporal Simultaneity

3.1 Definition of Double Fixed-Point Structure

Future fixed points alone are not sufficient to adequately describe the stability of the phenomena "present" and "past." We therefore introduce a *double fixed-point structure* in which future fixed points are coupled as a pair with fixed points on the present/past side.

Definition 3.1 (Double fixed point)**.** Consider a future fixed point $x^\star \in \mathcal{X}$ and a family of functions $G_t : \mathcal{X} \to \mathcal{X}$ generated from a present (or historical) state $x_t \in \mathcal{X}$. The pair $(x^\star, x_t)$ is called a *double fixed point* if

$$x^\star = \lim_{T\to\infty} F_T(x_t), \tag{1}$$

$$x_t = \lim_{s\to-\infty} G_s(x^\star) \tag{2}$$

hold simultaneously.

Equation (1) expresses convergence "from the present to the future," while equation (2) expresses a "reverse constraint from the future to the past." The simultaneous validity of both expresses that time is *closed as a double fixed point of future and past.*

3.2 Triple-Temporal Simultaneity

In FCT, past, present, and future are treated not as mere orderings, but as different projections of a single structure.

Definition 3.2 (Triple-temporal simultaneity)**.** Consider an abstract state space $\mathcal{S}$ with three maps

$$\pi_{\text{past}},\ \pi_{\text{present}},\ \pi_{\text{future}} : \mathcal{S} \to \mathcal{X}.$$

Triple-temporal simultaneity means that for some $\sigma \in \mathcal{S}$,

$$\pi_{\text{past}}(\sigma) = x_-, \quad \pi_{\text{present}}(\sigma) = x_0, \quad \pi_{\text{future}}(\sigma) = x_+$$

are simultaneously defined, and the triple (x_-, x_0, x_+) is compatible with the double fixed-point structure.

Intuitively, σ is an "informational state that simultaneously carries all three temporalities," and our observation manifests only through one particular projection π_{present}.

4 Semantic Density Tensor and Information Field

4.1 Information Field and Semantic Density

In many FCT papers, the world is treated as a "field on which a semantic density tensor is defined." Here we provide a minimal definition of this idea.

Definition 4.1 (Information field)**.** We call an abstract manifold (or topological space) $\mathcal{M}$ an *information field.* A point $p \in \mathcal{M}$ represents a "semantically distinguishable state."

Definition 4.2 (Semantic density scalar field)**.** We define a scalar field $\rho_{\text{sem}} : \mathcal{M} \times \mathbb{R} \to \mathbb{R}_{\geq 0}$ such that, for a point $p \in \mathcal{M}$ and time t, $\rho_{\text{sem}}(p, t)$ gives the "density of meaning concentrated at that point."

4.2 Semantic Density Tensor

To make spatial, temporal, and relational directions explicit, we introduce a tensor of semantic density gradients.

Definition 4.3 (Semantic density tensor)**.** When the information field $\mathcal{M}$ is locally described in a coordinate system $\{x^\mu\}$ $(\mu = 0, 1, \ldots, n)$, we define

$$I_{\mu\nu}(x,t) := \nabla_\mu \nabla_\nu \Psi_{\text{future}}(x,t)$$

and call it the *semantic density tensor*. Here, Ψ_{future} is the "future-oriented semantic potential," and ∇_μ is a covariant derivative based on the corresponding connection.

The intuitive meaning of $I_{\mu\nu}$ is that it simultaneously encodes:

- in which directions the "semantic gradient from the future" is strong;
- which structures are preferred in the direction of future fixed points;
- where creative allowance still remains.

4.3 Relation Between $\Lambda(T)$ and Semantic Density

We can link the future convergence function $\Lambda(T)$ and semantic density via a simple correspondence.

Definition 4.4 (Future convergence via semantic density)**.** For a point $x \in \mathcal{M}$ in the information field, we define

$$\Lambda(T)(x) := \frac{\exp\bigl(\alpha(T)\,\Psi_{\text{future}}(x,T)\bigr)}{Z(T)}, \qquad Z(T) = \int_{\mathcal{M}} \exp\bigl(\alpha(T)\,\Psi_{\text{future}}(y,T)\bigr)\,dy,$$

where $\alpha(T) \geq 0$ is a "convergence-strength parameter" that depends on the future scale.

If $\alpha(T) \to \infty$ as $T \to \infty$, then the maxima (or local maxima) of Ψ_{future} are extracted as future fixed points. Within this framework, the semantic density tensor $I_{\mu\nu}$ characterizes the local structure near the future fixed points.

5 Quantum Creative Allowance and Intermediate-Layer Structure

5.1 4D Future Information Field and 3D Manifestation

We now briefly redefine the structure treated in the extended FCT paper "Quantum Creation Regulation Theory" in a simplified form.

- $\Phi(T)$: four-dimensional future information field;
- $X(t)$: three-dimensional manifestation (state in ordinary physical space).

Rather than mapping one directly to the other, we introduce a quantum layer $\Psi(x,t)$ as an intermediate level:

$$X(t) = P\bigl(\Psi(x,t)\bigr), \qquad \Psi(x,t) \in Q\bigl(\Phi(T)\bigr), \tag{3}$$

where P is a three-dimensional projection operator and Q is a quantum admissibility regulation operator.

5.2 Definition of Quantum Creative Allowance

Between the higher-dimensional future information field and the lower-dimensional three-dimensional manifestation, there inevitably arises a "mapping allowance." We define this as the quantum creative allowance.

Definition 5.1 (Quantum creative allowance)**.** In the mapping from a future information field $\Phi(T)$ to a three-dimensional manifestation $X(t)$, we denote by

$$\mathcal{W} := \Delta_{\Phi \to X}$$

the total amount of structural degrees of freedom that are not preserved, and call this the *quantum creative allowance.*

Abstractly, we may also express it as the difference between the information content $H(\Phi)$ of Φ and the information content $H(X)$ of X:

$$\mathcal{W} = H(\Phi) - H(X),$$

where H is an appropriate entropy-like functional.

5.3 Uncertainty and Creative Allowance

The usual uncertainty relation

$$\Delta x \, \Delta p \gtrsim \frac{\hbar}{2}$$

is interpreted in the framework of FCT as a lower bound on the creative allowance $\mathcal{W}$, namely,

$$\Delta x \, \Delta p \;\geq\; f(\mathcal{W}),$$

where the "retention capacity of the future information field" imposes the constraint.

6 Abstract Formalization Toward the -System

6.1 Minimal Model of the -System

We represent the -System as an operator that receives three inputs: the future convergence function, the semantic density tensor, and the quantum creative allowance.

Definition 6.1 (Abstract -System)**.** We define an abstract -System as a map

$$\mathcal{L} : \big(\Lambda(T),\, I_{\mu\nu},\, \mathcal{W}\big) \mapsto \Pi_{\mathrm{act}},$$

where Π_{act} is a set of "projection operators onto the world" corresponding to actions, decisions, and structural updates.

Intuitively,

- $\Lambda(T)$: constraint from future fixed points (where convergence is to be directed);
- $I_{\mu\nu}$: local structure of semantic density (where meaning is dense);
- $\mathcal{W}$: available creative allowance (how far rewriting is possible);

are inputs, and the -System is a device that selects "which actions, decisions, and reprojections are admissible."

6.2 Mapping to Future-Causal Ethics and Social Design

In applications to ethics, social institutions, AI, and so forth, we select from Π_{act} those operators that minimize the deviation from the future fixed points.

Definition 6.2 (Future-causal selection principle)**.** Introduce an evaluation function $D_{\text{future}}(\Pi)$ as the "deviation from the future fixed points." The future-causal selection principle selects the operator $\Pi^\star$ via

$$\Pi^\star = \arg\min_{\Pi \in \Pi_{\text{act}}} D_{\text{future}}(\Pi).$$

On top of this framework, the following applied theories can each be positioned as projections:

- future-causal ethics (ethics as minimization of deviation from future fixed points);
- semantic government (governance as control of the configuration of semantic density);
- FCIR (international relations as interference of linguistic tensors between states);
- ENΛ (language design as deletion of operators that generate discrimination).

7 Finite-State Toy Model of the Future Convergence Function and Double Fixed Points

In this section, we concretize the future convergence function $\Lambda(T)$ and the double fixed-point structure, defined so far in an abstract way, as a simple model on a finite state space. The toy model presented here is not a direct description of physical reality. Its purpose is to make mathematically visible *how a structure in which states converge to a future fixed point can be implemented.*

7.1 State Space and Probability Simplex

Consider a finite set of states

$$\mathcal{X} = \{1, 2, \ldots, N\}.$$

We define the set of all probability distributions on this set as

$$\Delta_N = \left\{ p = (p_1, \ldots, p_N) \in \mathbb{R}^N \,\middle|\, p_i \geq 0,\ \sum_{i=1}^{N} p_i = 1 \right\}.$$

Equipped with the Euclidean distance

$$d(p, q) = \|p - q\|_2,$$

Δ_N is a complete metric space and a compact subset of $[0, 1]^N$.

We regard the "future-side target distribution" induced from the future information field as

$$q_T \in \Delta_N,$$

which corresponds to the discrete model of the "future fixed point" in FCT.

7.2 Future-Side Dynamics F_T and Present-Side Dynamics G_s

We define a map on Δ_N,

$$F_T : \Delta_N \to \Delta_N,$$

by

$$F_T(p) = (1-\alpha)\, p + \alpha\, q_T, \qquad 0 < \alpha < 1. \tag{4}$$

Intuitively, F_T is an operation that mixes the present distribution p toward the future distribution q_T via a "constraint from the future side."

Similarly, for the "past/present-side" update operation, we define

$$G_s : \Delta_N \to \Delta_N$$

by

$$G_s(p) = (1-\beta)\, p + \beta\, q_T, \qquad 0 < \beta < 1. \tag{5}$$

Both F_T and G_s share the same fixed point q_T, but the differences in α and β can be interpreted as differences in the strength of future dependence and the time scale of the updates.

The important point here is that we treat F_T and G_s not as maps with distinct origins, but as two dynamics that are *both constrained by the same future fixed point* q_T. This allows us to explicitly realize the double fixed-point structure as a finite-state model.

7.3 Contractivity of Both Maps and Existence of a Common Fixed Point

Lemma 7.1 (Contractive maps F_T and G_s)**.** *For any $p, q \in \Delta_N$,*

$$\|F_T(p) - F_T(q)\|_2 \le (1-\alpha)\, \|p-q\|_2,$$

$$\|G_s(p) - G_s(q)\|_2 \le (1-\beta)\, \|p-q\|_2$$

hold. Therefore, F_T and G_s are both contraction mappings on Δ_N.

Proof. From the definition (4),

$$F_T(p) - F_T(q) = (1-\alpha)(p-q),$$

and hence

$$\|F_T(p) - F_T(q)\|_2 = (1-\alpha)\, \|p-q\|_2.$$

Since $0 < \alpha < 1$, F_T is a contraction. The same argument using (5) shows that G_s is also a contraction. □

By the contraction mapping principle, we obtain the following.

Theorem 7.2 (Unique common fixed point as a future fixed point)**.** *Let Δ_N be the metric space defined above, and let F_T and G_s be defined by* (4) *and* (5)*, respectively. Then:*

1. *F_T has a unique fixed point $p^* \in \Delta_N$.*
2. *G_s has a unique fixed point $r^* \in \Delta_N$.*
3. *In this model, $p^* = r^* = q_T$ holds.*

Hence q_T is a common fixed point of F_T and G_s and gives a double fixed-point structure in the finite-state model.

Proof. By the contraction mapping principle, F_T has a unique fixed point p^*. From the fixed-point condition $F_T(p^*) = p^*$,

$$(1-\alpha)p^* + \alpha q_T = p^* \quad \Longrightarrow \quad \alpha(q_T - p^*) = 0 \quad \Longrightarrow \quad p^* = q_T.$$

Similarly, from $G_s(r^*) = r^*$ we obtain $r^* = q_T$. □

This toy model gives a rigorous realization, in a simple finite-state setting, of the FCT-inspired diagram

$$\text{future fixed point} \quad \leftrightarrow \quad \text{common fixed point of past/present dynamics.}$$

7.4 Interpretation of Dynamics at the Double Fixed Point

If we denote by

$$p^{(n)} = F_T^{\circ n}(p^{(0)})$$

the state obtained after n iterations of F_T, then

$$\lim_{n\to\infty} p^{(n)} = q_T$$

holds for any initial distribution $p^{(0)} \in \Delta_N$.

Likewise, iterations of G_s also converge to q_T. In this finite-state model, therefore,

- the future-side dynamics F_T, and
- the present-side dynamics G_s

are both *uniquely constrained by the same future fixed point q_T.*

This structure is a concrete mapping, in a finite-state probability model, of the abstract continuous version of the double fixed-point theory,

$$F_T(x^*) = x^*, \quad G_s(x^*) = x^*,$$

and provides the simplest prototype of "future-causal convergence" in FCT.

8 Entropy Representation of the Quantum Creative Allowance and a Toy Model

In this section, we concretize the "quantum creative allowance" $\mathcal{W}$ as a difference of information entropies, and show in a simple toy model that

$$\mathcal{W} \geq 0$$

holds. This serves as a mathematical underpinning for the statement that, in the projection from a future information field to a three-dimensional manifestation, *a creative allowance necessarily arises.*

8.1 Definition of Creative Allowance as an Entropy Difference

As in the previous section, consider a finite state space $\mathcal{X} = \{1, \ldots, N\}$. Let P_Φ be the distribution in the future information field, and P_X the distribution in the three-dimensional manifestation. We assume both are probability distributions on Δ_N.

Define Shannon entropy by

$$H(P) = -\sum_{i=1}^{N} P(i) \log P(i),$$

where the base of the logarithm is arbitrary (we use the natural logarithm below).

We then define the quantum creative allowance as

$$\mathcal{W} := H(P_\Phi) - H(P_X). \tag{6}$$

Intuitively,

- P_Φ: distribution of "potential states" in the higher-dimensional future information field;
- P_X: distribution after a subset of that potential has been projected to three dimensions;

and if $H(P_\Phi) \geq H(P_X)$, then

$$\mathcal{W} \geq 0$$

represents "the amount of information lost by projection, which can be reused for creation."

8.2 Monotonicity Theorem for Deterministic Projections

For simplicity, we consider the case where P_X is the image of P_Φ under a *deterministic map*. Let

$$f : \mathcal{X} \to \mathcal{Y},$$

with $Y = f(X)$, and define

$$P_X = \mathbb{P}(X = i), \quad P_Y = \mathbb{P}(Y = y).$$

If we identify $P_\Phi := P_X$ and $P_X := P_Y$, then the creative allowance becomes

$$\mathcal{W} = H(X) - H(Y).$$

Theorem 8.1 (Monotonicity of entropy under deterministic maps)**.** *For a discrete random variable X and its deterministic function $Y = f(X)$,*

$$H(Y) \leq H(X)$$

holds. Therefore, in the situation of (6)*,*

$$\mathcal{W} = H(P_\Phi) - H(P_X) \geq 0.$$

Proof. A basic fact of information theory is that the conditional entropy $H(X|Y) \geq 0$. By the chain rule,

$$H(X, Y) = H(Y) + H(X|Y),$$

and since $Y = f(X)$ is deterministic,

$$H(X, Y) = H(X).$$

Hence,

$$H(X) = H(Y) + H(X|Y) \geq H(Y),$$

i.e., $H(Y) \leq H(X)$. Translating back to the notation P_Φ, P_X yields the claim. □

This theorem shows that, as long as the three-dimensional manifestation distribution P_X is deterministically generated from the higher-dimensional future information field distribution P_Φ, *a non-negative creative allowance* $\mathcal{W} \geq 0$ *necessarily arises.*

In the language of FCT, this $\mathcal{W}$ functions as

"the residual degrees of freedom that can be used for convergence toward future fixed points,"

serving as the source of evolution, creation, and reconstruction.

8.3 Two-State Toy Model

As the simplest example, consider a two-state system $\mathcal{X} = \{0, 1\}$. Let the distribution in the future information field be

$$P_\Phi(0) = p, \quad P_\Phi(1) = 1 - p,$$

and consider an extreme model where, in the three-dimensional manifestation, we have

- no distinction between the two states (complete degeneracy).

Then

$$P_X(0) = 1$$

(all probability mass is compressed into a single state).

The entropies are

$$H(P_\Phi) = -p \log p - (1 - p) \log(1 - p),$$

$$H(P_X) = -1 \cdot \log 1 = 0.$$

Hence, the creative allowance is

$$\mathcal{W} = H(P_\Phi) - H(P_X) = -p \log p - (1 - p) \log(1 - p) \geq 0,$$

with equality holding only when $p \in \{0, 1\}$ (i.e., when the state is already fully determined).

Even this simple two-state model shows that the difference between

- the "diverse possibilities" P_Φ in the future information field, and
- the "determined state" P_X in three-dimensional manifestation

necessarily remains as a non-negative quantity $\mathcal{W}$. In FCT, this is interpreted as

creative allowance = semantic degrees of freedom still available to the future.

8.4 Reinterpretation of the Role of Creative Allowance in FCT

Through the finite-state model above, the inequality

$$\begin{aligned}\mathcal{W} &= H(\Phi) - H(X) \\ &\geq 0\end{aligned}$$

is understood not as a mere slogan, but as *an information-theoretically guaranteed "reserve of degrees of freedom."*

In FCT terminology:

- $H(\Phi)$: latent diversity retained by the information field in the vicinity of future fixed points;
- $H(X)$: diversity after the three-dimensional manifestation at the current moment has been determined;
- $\mathcal{W}$: creative degrees of freedom that can still be redistributed toward the future.

All forms of "change"—cosmic evolution, biological evolution, transformations of consciousness, renewal of civilization, and self-reconfiguration of AI—are positioned as consequences of

$$\mathcal{W} > 0.$$

If $\mathcal{W} = 0$ were to hold at the scale of the universe, the world would be completely frozen, and no newness could emerge: a "dead universe."

Thus,

$$\mathcal{W} > 0$$

can be read, in FCT, as

a necessary condition for the universe to continue creating.

The finite-state model and entropy representation presented in this section provide a minimal yet concrete mathematical underpinning for the intuitive FCT concept of the "quantum creative allowance."

9 Continuous-Time Model of Future Convergence Dynamics

Up to the previous section, we have given a discrete-time model on a finite state space $\mathcal{X} = \{1, \ldots, N\}$ for the future convergence function $\Lambda(T)$ and the double fixed-point structure, as well as an entropy representation of the quantum creative allowance $\mathcal{W}$.

In this section, we construct a toy model of *continuous time t and continuous state space* that is closer to physical intuition, and we formulate future convergence dynamics. The goal is to explicitly write, in the language of ordinary differential equations and stochastic differential equations, the FCT intuition of

"time evolution constrained by future fixed points."

9.1 State Space and Future Potential

We consider a continuous state $x \in \mathbb{R}^d$ as the state space. We model the constraint from the future side by a scalar field

$$\Phi_{\text{future}} : \mathbb{R}^d \to \mathbb{R}$$

(the future potential). A "future fixed point" in this setting is defined as a point $x^* \in \mathbb{R}^d$ satisfying

$$\nabla \Phi_{\text{future}}(x^*) = 0. \tag{7}$$

Moreover, when the Hessian

$$\nabla^2 \Phi_{\text{future}}(x^*) \succ 0$$

is locally positive definite around Φ_{future}, the point x^* is a local minimum of the future potential and corresponds, in continuous-state form, to a "future attractor" in FCT.

9.2 Deterministic Gradient Flow as Future Convergence

First, consider a noise-free deterministic time evolution

$$\dot{x}(t) = v(x(t)).$$

As an idealization where only constraints from the future side act, we define the velocity field by

$$\dot{x}(t) = -\nabla \Phi_{\text{future}}(x(t)). \tag{8}$$

This is the negative gradient flow of the potential Φ_{future}, and it satisfies the standard property

$$\frac{d}{dt}\Phi_{\text{future}}(x(t)) = \nabla \Phi_{\text{future}}(x(t)) \cdot \dot{x}(t) = -\|\nabla \Phi_{\text{future}}(x(t))\|^2 \leq 0. \tag{9}$$

Thus Φ_{future} is a Lyapunov function, and the orbit monotonically decreases the future potential over time.

Theorem 9.1 (Convergence to a local future fixed point)**.** *Suppose that Φ_{future} is C^2, x^* is an isolated critical point satisfying* (7)*, and the Hessian $\nabla^2 \Phi_{\text{future}}(x^*)$ is positive definite. Then any solution orbit $x(t)$ starting sufficiently close to x^* satisfies*

$$\lim_{t \to \infty} x(t) = x^*.$$

Proof. From the gradient flow (8) and monotonicity (9), $\Phi_{\text{future}}(x(t))$ is bounded and monotonically decreasing, and hence convergent. Moreover, $\frac{d}{dt}\Phi_{\text{future}}(x(t)) \to 0$ implies $\|\nabla \Phi_{\text{future}}(x(t))\| \to 0$. Under the regularity assumptions, standard arguments for stable equilibria ensure that $x(t)$ converges to x^*. □

This theorem shows that, in the continuous-time model,

"local minima of the future potential" = "stable fixed points of time evolution."

In FCT terms:

> Future fixed points are not the *result* of time evolution, but are embedded *from the outset as its destination.*

9.3 Continuous-Time Version of the Double Fixed-Point Structure

In the finite-state model of the previous section, we showed that the same distribution q_T is realized as a common fixed point of

- the future-side dynamics F_T, and
- the present-side dynamics G_s.

In the continuous-time setting, we introduce:

- the future potential Φ_{future}, and
- the present/observation-side potential U_{now}.

We consider the gradient flows associated with each:

Future side:

$$\dot{x}(t) = -\nabla\Phi_{\text{future}}(x(t)),$$

Present side:

$$\dot{x}(t) = -\nabla U_{\text{now}}(x(t)).$$

We assume that they share the same stable point x^*:

$$\nabla\Phi_{\text{future}}(x^*) = 0, \qquad \nabla U_{\text{now}}(x^*) = 0. \tag{10}$$

We then assume that the actual time evolution is given by a linear combination of the two:

$$\dot{x}(t) = -(1-\lambda)\,\nabla U_{\text{now}}(x(t)) - \lambda\,\nabla\Phi_{\text{future}}(x(t)), \qquad 0 < \lambda < 1. \tag{11}$$

Lemma 9.2 (Common fixed point in the continuous-time model)**.** *A point x^* satisfying* (10) *is a fixed point of the combined dynamics* (11)*.*

Proof. Substituting $x = x^*$ into (11), we obtain

$$\dot{x}(t) = -(1-\lambda)\,\nabla U_{\text{now}}(x^*) - \lambda\,\nabla\Phi_{\text{future}}(x^*) = 0.$$

□

Under appropriate convexity and positive-definiteness assumptions, x^* can become a stable equilibrium of the combined system. From the FCT perspective:

- U_{now}: "status quo potential" induced by past/present history;
- Φ_{future}: "future potential" corresponding to future fixed points;

and (11) is the simplest mathematical model of

continuous-time dynamics under double constraints from "past/present" and "future."

9.4 Stochastic Dynamics and Fokker–Planck Representation

We now introduce stochastic fluctuations representing quantum creative allowance and environmental noise. Consider the stochastic differential equation

$$dX_t = -\nabla\Phi_{\text{future}}(X_t)\,dt + \sqrt{2D}\,dW_t, \tag{12}$$

where $D > 0$ is a diffusion coefficient and W_t is a d-dimensional Brownian motion.

The probability density $\rho(x,t)$ corresponding to this SDE satisfies the Fokker–Planck equation

$$\frac{\partial}{\partial t}\rho(x,t) = \nabla\cdot\big(\rho(x,t)\,\nabla\Phi_{\text{future}}(x)\big) + D\,\Delta\rho(x,t). \tag{13}$$

The stationary distribution ρ_∞ satisfies

$$\frac{\partial}{\partial t}\rho_\infty = 0,$$

and thus

$$\nabla\cdot\big(\rho_\infty(x)\,\nabla\Phi_{\text{future}}(x)\big) + D\,\Delta\rho_\infty(x) = 0. \tag{14}$$

Under standard assumptions (regularity, boundary conditions, regular potential), one candidate stationary distribution is

$$\rho_\infty(x) \propto \exp\left(-\frac{\Phi_{\text{future}}(x)}{D}\right).$$

Here, D corresponds to the "thermal strength" of creative allowance; in the limit $D \to 0$, the deterministic gradient flow (8) is recovered.

From the FCT viewpoint:

- $\rho(x,t)$: "manifestation probability density" on the state space;
- ρ_∞: "future equilibrium distribution" consistent with the future potential.

9.5 Time Evolution of Entropy and Creative Allowance

We define the Shannon entropy of the probability density $\rho(x,t)$ by

$$H(\rho(t)) = -\int_{\mathbb{R}^d} \rho(x,t)\,\log\rho(x,t)\,dx.$$

Assuming that the entropy $H(\Phi)$ of the future information field remains constant, we can introduce the creative allowance

$$\mathcal{W}(t) = H(\Phi) - H(\rho(t)).$$

In general, the time evolution of $H(\rho(t))$ depends on the dynamics and boundary conditions and is not necessarily monotonic. However, if

- $\rho(t) \to \rho_\infty$ as $t \to \infty$, and
- $H(\rho_\infty) < H(\Phi)$,

then

$$\lim_{t\to\infty} \mathcal{W}(t) = H(\Phi) - H(\rho_\infty) > 0,$$

and *the creative allowance never completely disappears, even as time passes.*

In FCT language:

> The full potential of the future information field can never be perfectly mapped into three-dimensional manifestation, and some amount of "unused degrees of freedom" always remains. This residue functions as the source of new creation, evolution, and reconstruction.

9.6 Positioning of the Continuous-Time Toy Model

Relating the continuous-time model in this section to FCT structures yields roughly the following dictionary:

Concept in FCT	Correspondence in the continuous-time model
Future fixed point	Stable equilibrium x^* with $\nabla\Phi_{\text{future}}(x^*) = 0$
Double fixed-point structure	$\nabla\Phi_{\text{future}}(x^*) = \nabla U_{\text{now}}(x^*) = 0$
Future-causal dynamics	Gradient flow $\dot{x} = -\nabla\Phi_{\text{future}}(x)$
Creative allowance $\mathcal{W}$	Entropy difference $H(\Phi) - H(\rho(t))$
Future equilibrium	Stationary distribution $\rho_\infty \propto \exp(-\Phi_{\text{future}}/D)$

This continuous-time toy model is positioned as a "bridge" that:

- translates the abstract structure of FCT
- under minimal mathematical assumptions
- into a language shared by physics, stochastic processes, and information theory.

In a more refined theoretical development, one can extend this toy model to be consistent with the full FCT formalism using, for example:

- gradient flows on Riemannian manifolds;
- entropy gradient flows in Wasserstein distance;
- quantum Markov processes described by Lindblad equations.

However, since the aim of the present paper is to *explicitly specify a "mathematical prototype of future-causal dynamics,"* we leave these detailed generalizations for future work.

10 Comparison with Existing Theories and Positioning of the Present Framework

In this chapter, we clarify how the mathematical core of future convergence causal theory formulated in this paper relates to major existing frameworks—such as the principle of least action, gradient flows and optimal transport, Bayesian inference and information theory, teleology, quantum mechanics, and general relativity. Our aim is to show that:

1. the present theory is not a mere rephrasing of existing theories;
2. at the same time, it does not negate the effectiveness of existing theories;
3. rather, it is an "upwardly compatible framework" that includes and extends them.

In this chapter, we take as a representative core equation of future convergence causal theory the continuity equation

$$\frac{\partial \rho(x,t)}{\partial t} = -\nabla \cdot J(x,t), \tag{15}$$

$$J(x,t) = -D(\rho)\,\nabla\rho(x,t) - \rho(x,t)\,\nabla V_{\text{eff}}(x,t), \tag{16}$$

where V_{eff} is a linear combination of a past-dependent term V_{past} and a future-dependent term V_{future}:

$$V_{\text{eff}}(x,t) = \alpha\, V_{\text{past}}(x,t) + \beta\, V_{\text{future}}(x,t), \quad \alpha, \beta \geq 0, \tag{17}$$

and the limit $\beta \gg \alpha$ represents a situation in which "future causality dominates." The future potential V_{future} is given as a functional of the future information field Φ_{future} and of the set of future fixed points $\mathcal{F}$:

$$V_{\text{future}}(x,t) = \mathcal{V}\big(x,t;\, \Phi_{\text{future}}, \mathcal{F}\big). \tag{18}$$

10.1 Comparison with the Principle of Least Action and Variational Principles

In classical mechanics, the principle of least action fixes an initial state $(x(t_0), t_0)$ and a final state $(x(t_1), t_1)$, and demands that the path $x(t)$ connecting them extremize the action functional

$$S[x] = \int_{t_0}^{t_1} L\big(x(t), \dot{x}(t), t\big)\, dt. \tag{19}$$

Here, the future boundary condition $x(t_1)$ is mathematically specified, but is often interpreted physically as a condition that the observer sets after the fact.

By contrast, in the present theory, future causality is introduced in a fundamentally different way from the principle of least action, in that:

- future boundary conditions are not merely "boundary conditions for calculations";
- instead, they are given beforehand as an internal part of the system in the form of a "set of future fixed points $\mathcal{F}$."

Formally, the variational condition

$$\delta S[x] = 0 \tag{20}$$

and the continuum equation

$$\frac{\partial \rho}{\partial t} = \nabla \cdot \big(D(\rho)\,\nabla\rho + \rho\,\nabla V_{\text{eff}}\big) \tag{21}$$

are similar in the sense that both can be derived as extremal conditions of some functional. However,

1. the principle of least action primarily concerns the selection of a *single path*;
2. the present theory takes as its primary object the *time evolution of a distribution* $\rho(x,t)$ and its convergence to a set of future fixed points.

This difference is where the distinctiveness of the present theory lies.

10.2 Comparison with Gradient Flows and Optimal Transport

The structure used in this paper is formally close to gradient flows and the framework of optimal transport based on Wasserstein distance. For example, given a potential functional $\mathcal{F}[\rho]$, one often encounters the gradient flow equation

$$\partial_t \rho = \nabla \cdot \big(\rho \, \nabla \frac{\delta \mathcal{F}}{\delta \rho}\big). \tag{22}$$

Our theory extends this structure by writing

$$\mathcal{F}[\rho] = \mathcal{F}_{\text{past}}[\rho] + \mathcal{F}_{\text{future}}[\rho], \tag{23}$$

with a sum of a "past-side functional" and a "future-side functional", and furthermore letting $\mathcal{F}_{\text{future}}$ depend on the structure of the set of future fixed points $\mathcal{F}$.

The crucial point is that:

- the mathematical form may remain within the class of standard gradient flows;
- but the introduction of future causality fundamentally changes *what is being treated as the potential.*

In other words, our theory is not

> "a theory that proposes brand-new differential equations,"

but rather

> "a theory that provides new meanings and interpretations to existing equations."

10.3 Comparison with Bayesian Inference and Information Theory

In Bayesian inference and coding theory, quantities such as conditional probabilities and information measures are central, for example:

$$P(\theta \mid \text{data}) \propto P(\text{data} \mid \theta)\, P(\theta), \tag{24}$$

along with entropy H and mutual information I. These formulations typically condition on "data already obtained in the past."

In our theory, we introduce the creative allowance $\mathcal{W}$ as the quantity corresponding to information or uncertainty, and define it as

$$\mathcal{W}(t) = \mathcal{S}_{\text{future}}[\Phi_{\text{future}}] - \mathcal{S}_{\text{proj}}[\rho(\cdot, t)], \tag{25}$$

the difference between the entropy of the future information field and the entropy of the projected distribution at time t.

From this standpoint, the position of the present theory is that:

- Bayesian inference can be reinterpreted not as "optimal prediction given a fixed future," but
- as a special local approximation of future fixed points and creative allowance in regions where future causality is weak and creative allowance is small.

Thus, our theory does not negate information theory or Bayesian statistics, but positions them as

> "effective theories valid in regions where future causality is weak and creative allowance is small."

10.4 Difference from Teleology

Because the present theory posits that "future structure constrains the present," it may superficially resemble teleological frameworks. However, future causality here differs from classical teleology in that

$$\text{“purpose of a subject”} \neq \text{“future fixed point.”} \tag{26}$$

The set of future fixed points $\mathcal{F}$ has a structure that:

- is independent of the intentions and desires of individual subjects;
- is determined by the consistency and stability of the entire information field.

Thus, future causality in our theory does not mean that "someone's purpose guides the world," but rather that

> "consistency conditions of the entire information field result in the formation of fixed points on the future side, and these fixed points impose backward constraints on the present distribution."

10.5 Relation to Quantum Mechanics and General Relativity

We do not claim a direct unification with quantum mechanics and general relativity in this paper, as that would exceed its scope. However, at the level of mathematical structure, the following correspondences can be observed.

- In quantum mechanics, the time evolution of the wave function $\psi(x,t)$ is given by the Schrödinger equation
$$i\hbar\frac{\partial\psi}{\partial t} = \hat{H}\psi. \tag{27}$$
In the present theory, the probability current J of $\rho(x,t)$ is modified via the future potential V_{future} to describe an "effective non-unitary behavior."
- In general relativity, spacetime structure is locally determined by Einstein's equation
$$G_{\mu\nu} = 8\pi G\, T_{\mu\nu}. \tag{28}$$
Our theory suggests the possibility that the semantic density tensor ρ_{semantic} and the identity tensor $I_{\mu\nu}$ behave as an effective "future-side geometry."

At present, we do not provide a precise unified equation with these frameworks, but we emphasize that there is room to connect general relativity with

> "a geometry of information fields with future causality."

10.6 Uniqueness and Limitations of the Present Theory

Summarizing the above, the uniqueness of this mathematical core can be described as follows.

- Within the standard class of gradient-flow-type descriptions of $\rho(x,t)$, we explicitly introduce a future potential V_{future} and take convergence toward future fixed points as the primary object.

- By defining the creative allowance $\mathcal{W}$, we open a path to treating future causality and creativity within a single mathematical framework.
- We aim to consistently apply this framework to macroscopic structures such as consciousness, civilization, ethics, language, and international relations.

At the same time, the limitations of the present theory are:

1. the concrete form of the future potential V_{future} has not yet been fully specified;
2. experimental methods for measuring semantic density tensors and creative allowance remain at the proposal stage;
3. a full unification with quantum mechanics and general relativity remains unaddressed.

In light of these points, this mathematical core is positioned as

> "the first-generation (v1.0) mathematical skeleton of future convergence causal theory,"

and is explicitly presented as an "open theory" that presupposes future extension and criticism.

11 Observational Indicators and Testability

In this chapter, we summarize how the mathematical core of future convergence causal theory can *connect to the empirically observable world*, that is, to what extent it already contains a structure from the vantage point of testability. Although the theory spans many domains—cosmology, theories of consciousness, social theory, and so on—we confine ourselves here to presenting *prototype strategies of verification* based on the mathematical core.

11.1 Basic Strategy for Verification

The present theory treats seemingly abstract structures such as "future fixed points," "creative allowance," and "semantic density tensors." From the standpoint of testability, we adopt the following basic strategy.

1. We take the position that quantities not directly observable (future fixed points, semantic density, etc.) appear only as *statistical patterns of observable quantities.*
2. Testability is thus equivalent to asking whether

   > "statistical biases or asymmetries that are difficult to explain with standard present-causal models are naturally derived from this theory."

3. At the same time, the theory must possess "explicit falsifiable scenarios," i.e.,

   > if observational data take certain forms, then the theory is ruled out.

Below we present prototypes of observational indicators in four representative domains.

11.2 Asymmetry in Temporal Statistics: Detecting Future-Conditioned Statistics

The most direct arena for verification is the detection of "future-conditioned statistics" in time series data.

In a standard Markov process,

$$P(x_{t+1} \mid x_t, x_{t-1}, \dots) = P(x_{t+1} \mid x_t) \tag{29}$$

one typically assumes Markovian dependence only on the past.

From our perspective, if there is an influence from the set of future fixed points $\mathcal{F}$, then

$$P(x_t \mid x_{t-1}, x_{t-2}, \dots, \mathcal{F}) \neq P(x_t \mid x_{t-1}, x_{t-2}, \dots) \tag{30}$$

may display "anomalous future-conditioned statistics."

Experimentally, a verification strategy could be:

- prepare an environment where a "strong constraint" is introduced at some time t^* (e.g., rule changes, changes in objective functions);
- for data at $t < t^*$, test the difference between

$$P(x_t \mid \text{past}) \quad \text{and} \quad P(x_t \mid \text{past, conditions after } t^*) \tag{31}$$

  and evaluate their explanatory power.

If conditional probabilities that include "future-side constraints" consistently show statistically significant improvements in explanatory power, this can serve as an indicator of the "constraints from future fixed points" assumed in our theory.

11.3 Future-Dependent Indicators in Cognition and Consciousness

Regarding consciousness, intuition, and decision-making, the present theory includes the proposition that

> "the larger the creative allowance, the greater the inflow of future information."

Prototypes of observational indicators corresponding to this are:

1. **Dual-task experiments:** Compare "high creative-allowance states" (free association, meditative-like states) and "low creative-allowance states" (strong attentional constraints, strong time pressure). Change the conditions of future stimuli (rewards, feedback), and test whether statistical patterns of past choices change.
2. **Post-feedback relearning experiments:** After a task has been completed, provide "additional information from the future side," and examine whether there is a systematic bias in the *re-description of past events*. This interprets the reconfiguration of $\rho(x, t)$ according to the future potential as a macroscopic memory phenomenon.

These experiments remain at a conceptual level, but they suggest a way to statistically detect traces of future causality in terms of whether

> "changes of future conditions cause systematic changes in the frequency of past choices or in the structure of memory."

11.4 Application to Physical Experiments: Redesigned Delayed-Choice Experiments

Direct verification candidates in physics include quantum experiments such as delayed-choice and weak measurements. In our framework, the "indeterminacy" of quantum states is reinterpreted as creative allowance $\mathcal{W}$, leading to the following expectations.

- The overall future structure of the experimental protocol (e.g., data analysis policies and reporting methods) may introduce slight biases into the distribution of initial fluctuations.
- Not only the presence or absence of delayed choice, but also the "semantic context" (what purpose the choices serve) may influence the statistical distribution.

In an idealized form, if we could detect differences such as

$$P(\text{detection result} \mid \text{apparatus setting},\ \text{future analysis policy}) \neq$$

$$\text{P(detection result} \mid \text{apparatus setting)}, \quad (32)$$

they could be interpreted as candidates for "influences from future-side structure."

In practice, however, one must eliminate all biases and instrumental noise, and in this sense what we present here is only a *direction for experimental design.*

11.5 Reanalysis of Macro-Phenomenal Data: Societal and Civilizational Scale

For social phenomena and the history of civilization, the present theory offers the perspective that

> "the structure of future fixed points pulls history toward them."

To translate this into observational indicators, we must reanalyze long-term data series as follows.

1. For macro data such as economic, demographic, and technological indicators, extract the residual component after removing the variation explained by standard causal models.
2. Evaluate whether this residual structure is better explained by some "future-side constraints" (e.g., long-term goals, civilizational agendas).
3. If models that include future-side objectives consistently outperform standard models statistically, they can be seen as macro-scale traces of the future-causal structure assumed by the theory.

Verification at this level is highly long-term, and we do not engage in concrete data analysis in this paper. However, we explicitly emphasize the viewpoint of comparing

> "models that include future constraints vs. models that do not."

11.6 Falsifiability and Future Extensions of the Theory

Finally, we make clear in what sense this mathematical core is falsifiable. At minimum, the theory would be strongly constrained or ruled out if the following situations were consistently observed.

1. Even when future conditions are changed, no systematic changes in the statistical distributions of past and present can be detected at any scale.
2. In the transformation of consciousness and cognition, no correlation is observed between changes in degrees of freedom interpretable as "creative allowance" and the reconfiguration of semantic structure.
3. In macro-scale civilizational and social data, changes in long-term objectives and agendas do not yield any consistent improvement in explanatory power beyond existing causal models.

Conversely, the theory begins to gain empirical support only when

> "descriptions including future-side conditions repeatedly explain the real world better."

The observational indicators and verification strategies presented in this chapter are only a "first-generation proposal." Nonetheless, by making them explicit, future convergence causal theory is not merely a metaphysical claim, but is approaching the minimal criteria for a scientific theory in the sense that it:

- is formulated as a mathematical model;
- possesses candidate observational indicators;
- and has conditions for falsifiability.

Thus, the "mathematical core v1.0" constructed in this paper:

1. has its relation to existing theories clarified;
2. possesses a path toward empirical verification;
3. and serves as an open skeleton that accepts future extension and revision.

12 Conclusion and Directions for Future Extensions

In this paper, we have gathered the mathematical objects common to various papers in Future Convergence Theory (FCT) into the following five pillars:

1. the future convergence function $\Lambda(T)$ and future fixed points;
2. the double fixed-point structure and triple-temporal simultaneity;
3. the semantic density tensor $I_{\mu\nu}$ and the information field $\mathcal{M}$;
4. the quantum creative allowance $\mathcal{W}$ and the quantum layer as an intermediate level;

5. the abstract -System and the future-causal selection principle.

Through this, we have shown that:

- cosmology, quantum theory, theories of consciousness, AI, ethics, language, and international relations can all be relocated onto a single information field $\mathcal{M}$ and a semantic density tensor $I_{\mu\nu}$;
- quantum phenomena naturally arise as "creative allowance" from the dimensional gap between the future information field and the manifest world;
- the -System can be abstracted as a "future-causal decision device" that utilizes these structures.

Future tasks include:

1. providing rigorous mathematical conditions (existence theorems, uniqueness, stability) for each object;
2. constructing simple toy models (finite-state systems, examples on linear spaces) and explicitly computing $\Lambda(T)$ and $I_{\mu\nu}$;
3. designing concrete algorithms for "semantic density analysis" on existing data (physical experiments, language corpora, diplomatic documents, etc.).

This paper is the first attempt to bundle the "mathematical nucleus" of FCT into one place. Henceforth, applied papers will refer to the objects introduced here, thereby guaranteeing mutual consistency and reconstructability.

Acknowledgements
The content of this paper is the aggregation of reflections on the overall structure of Future Convergence Theory and related papers (including quantum civilization theory, quantum creation regulation theory, future-causal linguistics, future-causal international relations, semantic government, etc.).